

Freight Rate Prediction

Model evaluation report | October 2026

Executive summary

CatBoost was selected as the leading candidate using a chronological holdout. It achieved a holdout MAE of \$131.57. The selected model was then refit on all 48,000 labeled loads and used to produce the requested validation and December predictions.

1. Problem Overview

Predict the posted freight rate for each load from its lane, distance, equipment, weight, location coordinates, date, market conditions, and quote signal. The model is intended to estimate rates for future loads.

2. Data Exploration

Development data: 48,000 labeled loads with 14 columns. The separate final validation set contains 12,000 unlabeled loads with 13 columns. The target is posted_rate.

Predictors include pickup and delivery cities and coordinates, distance, equipment, weight, date, market_index, and quote_signal. Labeled dates span 2025-01-01 through 2025-10-31.

3. Data Quality Findings

Missing values: weight has 300 rows (0.625%); market_index has 374 rows (0.779%). Linear Regression and Random Forest impute numeric missing values with medians learned from their training data. CatBoost uses native missing-value handling.

There are 292 negative weights (minimum -47,500). In modeling copies, negative weight and its derived weight_per_mile are marked missing; source data is not overwritten. Investigate these records with the data owner before production use.

No full duplicate rows, non-positive distances, or non-positive target rates were found. Equipment has three categories: Dry Van, Reefer, and Flatbed. The target is right-skewed with high-end outlier candidates.

Methods and Validation

Feature construction, split design, and model configurations

4. Feature Engineering

Calendar features from date: month, day_of_week (Monday=0), ISO week, and quarter. The raw date and load_id are excluded from model inputs after feature extraction.

distance_squared captures a nonlinear distance effect. weight_per_mile is weight divided by positive distance; it is set missing when a negative weight is treated as invalid.

lane combines pickup and delivery into an origin-destination feature (pickup_delivery). pickup, delivery, equipment, and lane are categorical predictors; missing categorical values use a __MISSING__ marker.

Original numeric fields, including coordinates, distance, weight, market_index, and quote_signal, remain model features. CatBoost consumes categories natively; the linear and Random Forest pipelines one-hot encode them.

5. Validation Strategy

A chronological holdout approximates forecasting into the future and avoids training on dates later than those evaluated. The 304 unique labeled dates were split at 2025-09-01, keeping all loads for any date in the same partition.

Training partition: 38,477 loads dated 2025-01-01 to 2025-08-31. Holdout: 9,523 loads dated 2025-09-01 to 2025-10-31. Preprocessing was fit on the training partition only.

The separate 12,000-row validation set is unlabeled (2025-11-01 to 2025-12-31), so it was reserved for prediction and not used to calculate model-selection metrics.

6. Models Evaluated

Linear Regression: numeric median imputation and one-hot encoding as a simple baseline.

Random Forest: 300 trees, max_features=0.5, minimum leaf size 2, fixed random seed; numeric median imputation and one-hot encoding.

CatBoost: 800 iterations, depth 8, learning rate 0.05, RMSE objective, fixed random seed; native categorical handling for pickup, delivery, equipment, and lane.

Results and Model Selection

All metrics below are from the same Sep-Oct 2025 chronological holdout

7. Results

Lower values indicate smaller errors. MAE is the primary selection metric; RMSE emphasizes large errors.

Model	MAE (\$)	RMSE (\$)	MAPE
Linear Regression	\$187.14	\$649.07	8.37%
Random Forest	\$139.19	\$655.53	6.07%
CatBoost	\$131.57	\$642.17	6.12%

8. Final Model Selection

CatBoost was selected as the leading candidate because it had the lowest holdout MAE (\$131.57) and the lowest RMSE (642.17). *Its MAE improved by 55.57 (29.7%) versus Linear Regression and by \$7.62 (5.5%) versus Random Forest.*

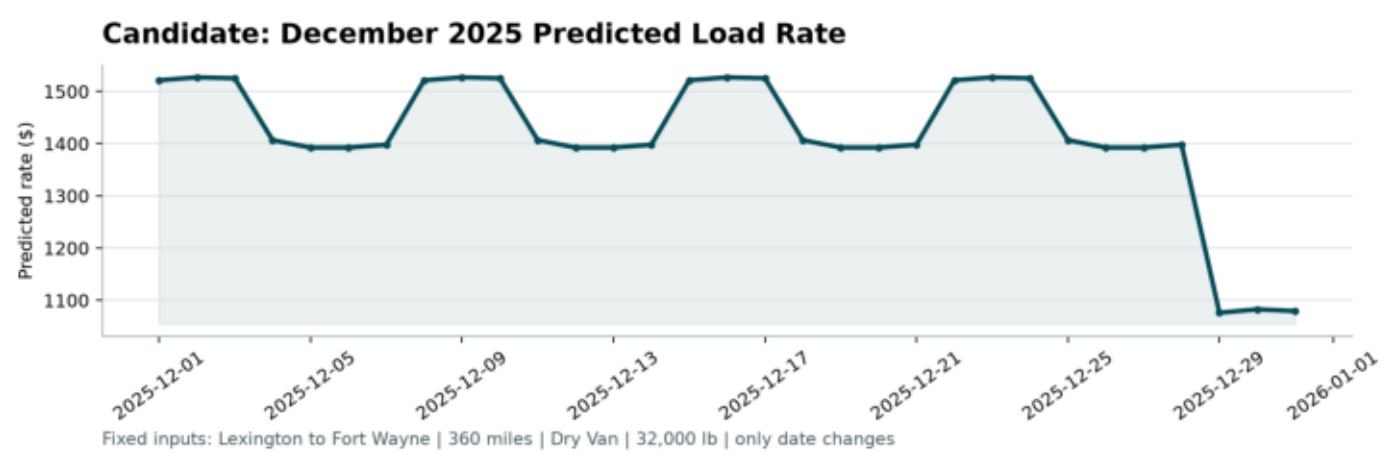
Random Forest had a marginally lower MAPE (6.07% vs. CatBoost 6.12%); selection prioritized MAE as specified in the validation plan. These are initial settings and a single holdout comparison, not a tuned or multi-period guarantee.

After comparison, CatBoost was refit on all 48,000 labeled development rows. That full-data model generated predictions for the 12,000 unlabeled validation loads and the December chart inputs. The unlabeled predictions are not scored in this report.

Recommended next step: rolling-origin evaluation across multiple historical cutoffs before production deployment.

December Prediction Chart

9. December Prediction Chart



The scorer validated one predicted rate for each date from December 1–31, 2025. Pickup, delivery, distance, equipment, and weight remain fixed; only the date changes. The chart is a model-generated candidate profile, not observed market truth.